

REFNO:074879

Roll No. _____

B.Sc. (Hons.) SEMESTER IV EXAMINATION 2023-24
MATHEMATICS
MTB - 401 : Partial Differential Equations

Time : Three hours

Max. Marks : 70

(WRITE YOUR ROLL NO. AT THE TOP IMMEDIATELY ON THE RECEIPT OF THIS QUESTION PAPER)

The figures in the right hand margin indicate marks.

Answer five questions including Question 1 which is compulsory.

1. Attempt all parts of this question

- (a) Form the partial differential equation by eliminating h and k from the equation:

$$(x - h)^2 + (y - k)^2 + z^2 = \lambda^2$$

[2]

- (b) Form a partial differential equation by eliminating arbitrary functions f and g from the equation: $y = f(x + \alpha t) + g(x - \alpha t)$. [2]

- (c) Solve the p.d.e. $z \frac{\partial z}{\partial x} = -y$. [2]

- (d) Find the complete integral of $z = px + qy + p^2 + q^2$, where $p = \partial z / \partial x$, $q = \partial z / \partial y$. [2]

- (e) Solve $(D_x^3 - 4D_x^2 D_y + 4D_x D_y^2)z = 0$, where $D_x \equiv \partial / \partial x$, $D_y \equiv \partial / \partial y$. [3]

- (f) Find the general solution of $r + p - q = z$. [3]

- (g) Solve $x^2 r - y^2 t + xp - yq = 0$ by reducing it in linear partial differential equation with constant coefficients. [3]

- (h) Solve $xr + p = 9x^2 y^2$. [3]

- (i) Check whether the partial differential equation $3r + 6s + 4t = 0$ is elliptic. [2]

2. (a) Obtain the general integral of the partial differential equation $p + 3q = 5z + \tan(y - 3x)$. [5]

- (b) Apply Charpit's method to find the complete integral of the equation $px + qy + z = zq^2$. [7]

3. (a) Find the general integral of the equation $r + s - 6t = y \cos x$. [5]

- (b) Reduce the equation

$$x^2 \frac{\partial^2 z}{\partial x^2} - 4xy \frac{\partial^2 z}{\partial x \partial y} + 4y^2 \frac{\partial^2 z}{\partial y^2} + 6y \frac{\partial z}{\partial y} = x^2 y^3$$

to linear equation with constant coefficients and then solve it. [7]

(2)

4. (a) Solve $(2D_x^2 - 5D_xD_y + 2D_y^2)z = 24(y - x)$. [5]

- (b) Solve the non-homogeneous partial differential equation

$$(D_x^2 - D_xD_y - 2D_y^2 + 2D_x + 2D_y)z = e^{2x+3y}.$$

[7]

5. (a) Solve: $yz - p = x^2y^2 \cos(xy)$. [5]

- (b) Reduce the following equation to canonical form and solve it

$$y^2 \frac{\partial^2 z}{\partial x^2} - 2xy \frac{\partial^2 z}{\partial x \partial y} + x^2 \frac{\partial^2 z}{\partial y^2} = \frac{y^2}{x} \frac{\partial z}{\partial x} + \frac{x^2}{y} \frac{\partial z}{\partial y}.$$

[7]

6. (a) Solve the following partial differential equation by Monge's method:
 $y^2r - 2yzs + t = p + 6y$. [5]

- (b) Solve: $r + 4s + t + rt - s^2 = 2$. [7]

7. (a) Use the method of separation of variables to solve the heat equation

$$\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < 1, \quad t > 0,$$

subject to the boundary conditions $u(x, 0) = x^2 - x$, $u(0, t) = u(1, t) = 0$. [6]

- (b) Solve the wave equation

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

using the method of separation of variables. [6]

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